

Contrastive Time-Series Anomaly Detection

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Abstract—In addition to its success in representation learning, contrastive learning is effective in image anomaly detection. Although contrastive learning depends significantly on data augmentation methods, time-series data augmentation for time-series anomaly detection is not investigated sufficiently. Additionally, although time-series data share a temporal context, the existing contrastive loss contrasts temporally related samples, in which deteriorated anomaly detection performance is observed on time-series data. Herein, we propose contrastive multivariate time-series anomaly detection (CTAD), a multivariate time-series anomaly detection framework that addresses these challenges by incorporating a one-class learning scheme into the contrastive loss based on meticulously designed time-series data augmentations. Specifically, we propose seven types of general time-series data augmentations to be applied variable- and point-wise, and provide guidance on data augmentation methods for contrastive time-series anomaly detection. The superiority of the one-class contrastive loss and the appropriate selection of time-series data augmentation allow CTAD to achieve outstanding performance in multiple datasets, even using a simple long short-term memory network. Furthermore, CTAD is robust to noise as it trains a noise-invariant network. This enables up to $47\times$ faster and $20\times$ more memory-efficient anomaly detection performance compared with existing methods while affording robustness, which are essential considerations in real-world applications.

Index Terms—Anomaly detection, contrastive learning, multivariate time-series, one-class classification.

I. INTRODUCTION

ANOMALY detection aims to detect anomalous behaviors that deviate from normal behaviors. As cyber-physical systems (CPS) become more complex, more sensors and actuators are used to operate the system [1], [2], [3]. This complexity renders anomaly detection in CPS more challenging, and unexpected behaviors can result in monetary damage or total system

failure. Advances in deep learning methods for anomaly detection have enabled accurate detections of anomalies in multivariate time-series [4], [5], [6], [7]. Moreover, deep-learning-based anomaly detection has shown remarkable adaptability across various practical scenarios, including medical diagnosis [8], gesture recognition [9], visitation prediction [10], and other applications [11], [12]. Nonetheless, a more accurate, fast, and robust multivariate time-series anomaly detection framework that is applicable to real-world scenarios is necessitated.

Contrastive learning has received attention owing to its anomaly detection performance in the field of computer vision [13], [14], [15]. For an encoder trained using contrastive learning, unseen anomalies or out-of-distribution samples are assumed to be encoded at a distance from the normal samples in the embedding space. Thus, the anomaly score for an unseen sample is defined based on the distance between the unseen sample and training samples, which are normal or in-distribution samples. Existing contrastive learning-based anomaly detection methods in the image domain rely on instance discrimination by exploiting contrastive loss.

Owing to the successful incorporation of data augmentation into contrastive learning frameworks, data augmentation has become the core component of contrastive learning frameworks. Existing approaches are typically based on instance discrimination [16], where representations from different instances are discriminated in the embedding space. However, instance discrimination can degrade the representation quality because semantically related samples (e.g., multiple dog images) in a mini-batch or queue can be regarded as negatives [17], [18].

In this study, we focus on some challenges in applying an instance discrimination-based contrastive learning framework to unsupervised multivariate time-series anomaly detection. First, for multivariate time-series data, one continuous signal is segmented with a sliding window, and each segment serves as an instance. Therefore, a temporal correlation exists between different instances, and treating them as negatives separates the representations of related instances. Second, suitable data augmentation for contrastive time-series anomaly detection must be investigated. Although data augmentations in image contrastive learning (e.g., cropping, color jittering, and rotation) have been investigated [13], [19], these spatial or image-specific augmentations are not applicable to time-series data. Because the appropriate data augmentation method is vital to anomaly detection performance [19], [20], extensive research for determining the optimal composition is indispensable.

A contrastive learning framework for multivariate time series anomaly detection, called contrastive multivariate time-series anomaly detection (CTAD), is proposed herein. CTAD differs

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from previous contrastive methods for anomaly detection in that the contrastive learning loss is based on one-class discrimination instead of instance discrimination. The proposed one-class discrimination considers all instances within the training distribution as positives, whereas the negatives are sampled solely from the augmented out-of-distribution. To contrast the negatives, CTAD generates synthetic anomalies through negative data augmentation, which intentionally generates out-of-distribution data and has been effectively leveraged in the image domain [13], [15], [21]. We categorize time-series data augmentations into three types: scale perturbation, temporality perturbation, and noise interference. We empirically show that scale-perturbing augmentations are effective as negative data augmentation for contrastive time-series anomaly detection.

CTAD exhibits remarkable performance even with a simple encoder architecture. With one layer LSTM [22], followed by projection layers consisting of two multi-layer perceptron (MLP) layers with batch normalization, CTAD outperforms other alternatives that adopt complex architectures. Its simple network enables fast and memory-efficient inferences. Our main contributions are summarized as follows:

- To the best of our knowledge, this is the first study to adopt one-class learning to a contrastive learning framework for multivariate time-series anomaly detection. We propose a one-class contrastive learning loss scheme that combines contrastive learning loss and a one-class learning scheme.
- We propose a time-series data augmentation strategy for contrastive time-series anomaly detection that is generalizable, generates various samples by increasing randomness, and generates hard-negative samples.
- We inspect seven general time-series data augmentation methods and provide a guideline for selecting the appropriate data augmentation method in contrastive time-series anomaly detection, based on experiments and ablation studies.
- We empirically demonstrate the validity of our proposed framework using multiple real-world datasets and show that it outperforms the compared methods even with a simple LSTM-based architecture.

II. RELATED WORK

A. Contrastive Learning

Contrastive learning is a self-supervised representation learning framework that encodes the semantic information of data without hand-labeled supervision. It causes the representation of semantically related samples (i.e., positives) to be attracted while being repelled from the others (i.e., negatives). Recent works define a positive sample as an augmented view from the same instance (e.g., image) and define a negative sample as an augmented view from all other instances in a mini-batch [19] or dictionary queue [23]. Recent contrastive learning methods depend significantly on data augmentation to generate semantically related samples [19], [20], [23]. Determining effective augmentation is critical to the performance of contrastive learning and depends on the data domain. For instance, color jittering is applicable to image data, but not applicable to time-series data.

B. Contrastive Learning for Time-Series

Owing to the superior performance of contrastive learning in image representation learning across many downstream tasks and multiple domains [24], [25], researchers have leveraged the contrastive learning framework for time-series representation learning. SeqCLR [26] proposed channel recombination preprocessing and time-series data augmentations for the contrastive representation learning of electroencephalogram (EEG) signals. However, this approach is only meaningful for bio-signals, such as EEG signals, and data augmentation is particularly suitable for EEG signals. CLOCS [27] learns a patient-specific representation of electrocardiogram (ECG) signals via contrastive learning. In this regard, a data augmentation method that considers domain-specific knowledge of ECG signals is designed. The transformed ECG segments originating from the same patient are regarded as a positive pair. GATE [28] proposed a self-supervised learning method to classify neuro-disease using functional magnetic resonance imaging (fMRI). To extract informative features from fMRI, GATE uses a domain-specific augmentation strategy to effectively augment the population graph, which is an input to the graph convolutional network. Similar to the image domain, appropriately designed data augmentation is crucial to the contrastive learning of time-series data. However, existing data augmentation strategies typically require data-specific knowledge [26], [27], [28], rendering them incompatible with other time-series data. Therefore, we aim to identify the general time-series data augmentation methods for contrastive learning that do not require domain-specific background knowledge. Moreover, we focus on effective data augmentations for time-series anomaly detection instead of representation learning.

C. Contrastive Learning for Anomaly Detection

Contrastive learning demonstrates high representation quality that is applicable to many downstream tasks and transfer learning [29]; recently, it has been shown to be effective in detecting out-of-distribution data in the image domain [13], [14], [15].

CSI [13] exploits undesirable augmentations for standard contrastive representation learning (e.g., rotations). It additionally generates instances from those augmentations and trains the encoder with existing instance discrimination-based contrastive loss and an auxiliary classifier to predict data augmentation. SSD [14] proposed an anomaly score based on the cluster-conditioned Mahalanobis distance in the embedding space; however, the encoder network was trained with the existing instance discrimination-based contrastive loss. In another study [15], a one-class classifier was trained on a learned representation in a two-stage manner. This is different from our study in that we trained the one-class contrastive loss in an end-to-end manner.

These contrastive anomaly detection approaches are typically based on instance-discrimination-based contrastive learning. However, instance discrimination is unsuitable for time-series anomaly detection because it contrasts with temporally related data. We address this concern by proposing one-class contrastive loss, which is introduced in the following section. Table I

TABLE I
SUMMARY AND COMPARISON OF RELATED WORKS AND OUR WORK

Method	Data domain	Domain knowledge	Contrastive loss	Goal
SimCLR [19]	image	✗	instance discrimination	representation learning
CSI [13]	image	✗	instance discrimination	novelty detection
SSD [14]	image	✗	instance discrimination	novelty detection
SeqCLR [26]	time-series	✓	instance discrimination	representation learning
CLOCS [27]	time-series	✓	supervised contrastive	representation learning
GATE [28]	time-series	✓	non-contrastive	representation learning
CTAD (ours)	time-series	✗	one-class contrastive	anomaly detection

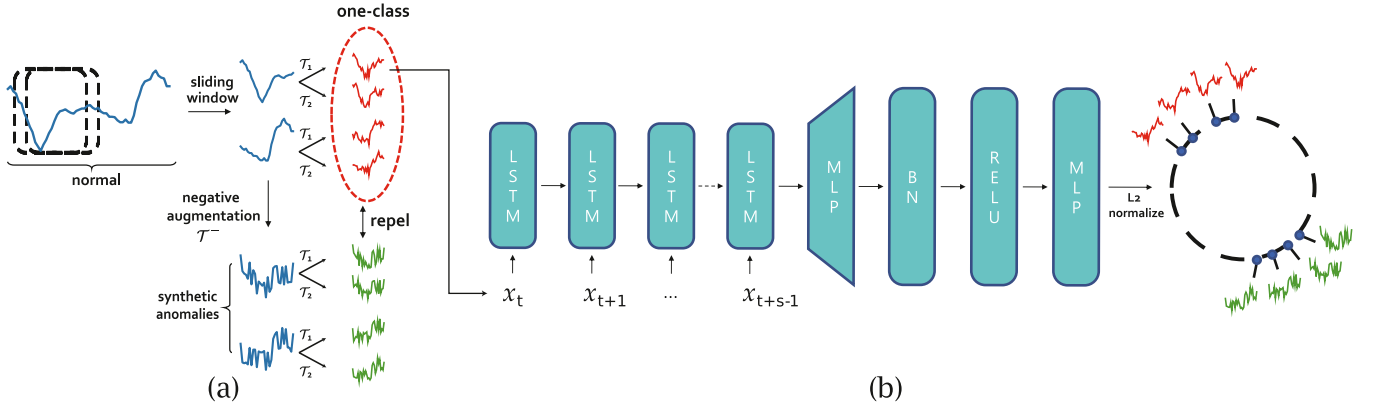


Fig. 1. Overview of CTAD framework. (a) Construction of window-sliced batch and generation of synthetic anomalies via negative augmentation. (b) Encoder network and schematic of proposed one-class contrastive loss.

summarizes methods associated with our proposed contrastive time-series anomaly detection framework.

III. METHODS

Fig. 1 illustrates the overall training process of the CTAD framework. CTAD attracts all window samples augmented from the training data because they share temporal context and repels synthetic anomalies, which are generated via an augmentation function named *negative augmentation*. In the following subsections, each component of CTAD is elaborated. In Section III-A, we first formulate unsupervised multivariate time-series anomaly detection problems and notations. In Section III-B, we introduce a contrastive representation and its limitations in time-series anomaly detection. In Section III-C, we explain the inspected time-series data augmentations for CTAD. Finally, we introduce a one-class contrastive loss that complements the limitations of the existing contrastive loss in Section III-D and an anomaly score in Section III-E.

A. Problem Formulation

In unsupervised multivariate time-series anomaly detection, the unlabeled train dataset $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\} := \{\mathbf{x}_t\}_{t=1}^T$ is assumed to be composed of normal observations for T time steps. It should be noted that $\mathbf{x}_t \in \mathbb{R}^d$, where d denotes the number of variables. During training, a contiguous series $\mathbf{X}$ is sliced

with a window size s to construct the window-sliced batch $\mathbf{W} = \{\mathbf{W}_t\}_{t=1}^{T-s+1}$, where $\mathbf{W}_t = \{\mathbf{x}_t, \mathbf{x}_{t+1}, \dots, \mathbf{x}_{t+s-1}\} \in \mathbb{R}^{d \times s}$.

At the inference time, a window-sliced test batch $\mathbf{W}' = \{(\mathbf{W}'_t, y'_t)\}_{t=1}^{T'-s+1}$ is constructed from the labeled test dataset $(\mathbf{X}', \mathbf{Y}') = \{(\mathbf{x}'_t, y'_t)\}_{t=1}^{T'}$. The test dataset is composed of normal and anomalous observations for T' time steps, and a binary label $y'_t \in \{0, 1\}$ indicates whether $\mathbf{x}'_t$ is an anomalous observation ($y'_t = 1$) or not ($y'_t = 0$). An anomaly score $\mathcal{S}$ is defined for a specified unseen window $\mathbf{W}'_t$. If the anomaly score is above the selected threshold, then the observation at the final time step of the window is predicted to be an anomaly ($\hat{y}_{t+s-1} = 1$).

B. Contrastive Representation Learning

Using stochastic data augmentation, we generate two semantically related “views” of the window segments. First, from $\mathbf{W}$, mini-batch $\mathcal{B} = \{\mathbf{W}_t\}_{t=0}^{N-1}$ with batch size N is constructed. Inspired by recent representative works on contrastive learning [19], [23], we introduce an augmentation random variable $\mathcal{A}$ whose state space is a set of augmentation functions. Using two augmentation functions $\mathcal{T}_1$ and $\mathcal{T}_2$, which are the realizations of augmentation random variable $\mathcal{A}$, a multi-viewed mini-batch $\tilde{\mathcal{B}}$ is constructed as follows:

$$\tilde{\mathcal{B}} = \{\tilde{\mathbf{W}}_t\}_{t=0}^{2N-1},$$

$$\{\tilde{\mathbf{W}}_t\}_{t=0}^{N-1} = \mathcal{T}_1(\mathcal{B}), \{\tilde{\mathbf{W}}_t\}_{t=N}^{2N-1} = \mathcal{T}_2(\mathcal{B}) \quad (1)$$

where $\mathcal{T}_1(\mathcal{B})$ is a set of augmented samples obtained by applying $\mathcal{T}_1$ on each element of mini-batch $\mathcal{B}$, i.e., $\mathcal{T}_1(\mathcal{B}) := \{\mathcal{T}_1(\mathbf{W}_t)\}_{t=0}^{N-1}$.

Before introducing the proposed one-class contrastive loss, we first elaborate on the contrastive loss of SimCLR [19] to highlight its limitations in time-series anomaly detection. SimCLR follows instance discrimination [16], in which a view originating from the same instance is considered *positive*, whereas views from other instances are considered *negatives*. Based on the notations above and the formulation of instance discrimination, one can define the index of the positive sample for the anchor $\tilde{\mathbf{W}}_i$ as $i^+ = (i + N) \bmod 2N$, and the index set of the negative samples as $i^- = \{0, 1, \dots, 2N - 1\} \setminus \{i, i^+\}$. It should be noted that one positive sample exists, and all the other $2N - 2$ samples except for the anchor are negative samples.

Let $\mathbf{z}_i = f_\theta(\tilde{\mathbf{W}}_i)$ be the encoded vector of $\tilde{\mathbf{W}}_i$ by encoder f_θ , and $\text{sim}(\mathbf{z}_i, \mathbf{z}_j)$ be the cosine similarity between two vectors $\mathbf{z}_i$ and $\mathbf{z}_j$:

$$\text{sim}(\mathbf{z}_i, \mathbf{z}_j) = \frac{\mathbf{z}_i^\top \mathbf{z}_j}{\|\mathbf{z}_i\| \|\mathbf{z}_j\|}. \quad (2)$$

The contrastive loss of SimCLR, which is termed as NT-Xent (the normalized temperature-scaled cross entropy loss), is expressed as follows:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=0}^{2N-1} \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{i^+})/\tau)}{\sum_{j \in i^- \cup \{i^+\}} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}, \quad (3)$$

where τ is the temperature-scaling hyper-parameter. When NT-Xent is used as a training objective, the similarity of the encoded vector between the anchor and positive sample increases (maximizing the numerator inside the logarithm). By contrast, the similarities between the anchor and negative samples decrease (minimizing the denominator inside the logarithm).

NT-Xent is based on instance discrimination; therefore, only one positive sample exists for each instance. However, because of the temporal characteristics of the time-series data, other semantically related samples within the same mini-batch can exist. Thus, regarding these samples as negatives generates false negatives; they degrade the quality of the learned representation [17], which can be more severe in time-series data that share temporal relations. Hence, we first devise time-series data augmentation methods and then propose a one-class contrastive loss based on the proposed time-series data augmentation methods.

C. Time-Series Data Augmentation

In this subsection, we propose a general time-series data augmentation method to generate positive and negative samples. We inspected seven general time-series data augmentation methods to generate semantic-preserving views and synthetic anomalies: amplification, additive bias, dropout, permutation, masking, reversing, and additive Gaussian noise. We categorized these time-series data augmentations into three types based on the augmentation characteristics: scale perturbation, temporality perturbation, and noise interference. Fig. 2 and Algorithm 1 show each type of time-series data augmentation.

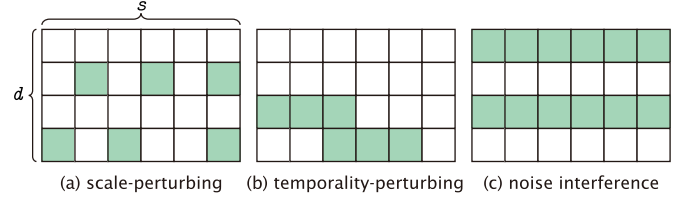


Fig. 2. Three types of time-series data augmentations. Colored squares represent randomly selected observations to be augmented. In all three cases, $s = 6$, $d = 4$, and $n_v = 2$. It should be noted that (c) does not require p , whereas $p = 0.5$ in both (a) and (b). Best visualized in color.

Scale-perturbing data augmentation changes the scale of the time-series. We designed these augmentations such that they operate point-wise. For a specified $\mathbf{W}_t$, the variables and time steps to apply augmentation were randomly and independently selected (lines 2–5). For the selected points across the selected variables, the deterministic point-wise operation was applied to augment $\mathbf{W}_t$ (line 7).

Scale-perturbing data augmentation comprises amplifying ($\mathcal{T}_{\text{amplify}}$), biasing ($\mathcal{T}_{\text{bias}}$), and dropping out ($\mathcal{T}_{\text{dropout}}$) time series, as follows:

$$\mathcal{T}_{\text{amplify}}(x_t; a) = a \times x_t \quad (4)$$

$$\mathcal{T}_{\text{bias}}(x_t; b) = x_t + b \quad (5)$$

$$\mathcal{T}_{\text{dropout}}(x_t) = 0. \quad (6)$$

The diversity and strength of the augmentations are controlled by two parameters: the number of variables n_v to select, and the percent p of selected points per variable. In the case of amplification and additive bias, the augmentation strength can also be represented by an amplification factor a and a bias scale b .

Temporality perturbing data augmentations use a continuous univariate time-series as input to change the temporal information of the time-series. These include reversing ($\mathcal{T}_{\text{reverse}}$), permuting, ($\mathcal{T}_{\text{permute}}$) or masking ($\mathcal{T}_{\text{mask}}$) time-series:

$$\mathcal{T}_{\text{reverse}}([x_1, \dots, x_t]) = [x_t, \dots, x_1] \quad (7)$$

$$\mathcal{T}_{\text{permute}}([x_1, \dots, x_t]) = \pi([x_1, \dots, x_t]) \quad (8)$$

$$\mathcal{T}_{\text{mask}}([x_1, \dots, x_t]) = [0, \dots, 0]. \quad (9)$$

The temporality perturbing augmentations involve two parameters, n_v and p , as well as scale-perturbing augmentations. For a multivariate time-series to be augmented, n_v variables are randomly selected. For each selected variable, continuous $t = \lfloor s \times p \rfloor$ time steps are selected from a randomly selected starting point.

Noise interference data augmentation augments the time-series by injecting additive white Gaussian noise, which is independent and identically distributed Gaussian noise with zero mean and variance σ^2 :

$$\mathcal{T}_{\text{Gaussian}}([x_1, \dots, x_t]; \sigma) = [x_1, \dots, x_t] + \mathbf{w},$$

where $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. (10)

The interference is applied in a variable-wise manner using n_v . This simple variation diversifies randomness because $\binom{d}{n_v}$

Algorithm 1: Time-Series Data Augmentation.

Input: $\mathbf{W}_t \in \mathbb{R}^{d \times s}$, $\mathcal{T}$, n_v , p
Output: $\mathbf{W}_t^{\text{aug}}$ (the augmented version of $\mathbf{W}_t$)

```

1  $\mathbf{W}_t^{\text{aug}} \leftarrow \mathbf{W}_t$ 
2  $\{\text{variables}\} \leftarrow \text{SelectVariables}(d, n_v)$ 
3 if  $\mathcal{T}$  is scale-perturbing then
4   for  $i \in \{\text{variables}\}$  do
5      $\{\text{points}\} \leftarrow \text{SelectPoints}(s, p)$ 
6     for  $j \in \{\text{points}\}$  do
7        $\mathbf{W}_t^{\text{aug}}[i][j] \leftarrow \mathcal{T}(\mathbf{W}_t[i][j])$  // point-wise
8 else if  $\mathcal{T}$  is temporality-perturbing then
9   for  $i \in \{\text{variables}\}$  do
10     $l, r \leftarrow \text{SelectPeriods}(s, p)$ 
11     $\mathbf{W}_t^{\text{aug}}[i][l:r] \leftarrow \mathcal{T}(\mathbf{W}_t[i][l:r])$ 
12 else if  $\mathcal{T}$  is noise-interference then
13   for  $i \in \{\text{variables}\}$  do
14     $\mathbf{W}_t^{\text{aug}}[i] \leftarrow \mathcal{T}(\mathbf{W}_t[i])$ 

```

TABLE II
PERFORMANCE COMPARISON OF CTAD WITH NT-XENT

Contrastive loss	AUROC	AUPR	F1
NT-Xent	0.465	0.127	0.279
CTAD	0.880	0.802	0.841

variable combinations are possible. Moreover, the strength of the augmentation can be controlled not only by σ , but also n_v .

D. One-Class Contrastive Loss

In this subsection, we describe the process to exploit the proposed time-series data augmentations and address the limitations of the existing contrastive loss in time-series anomaly detection using a one-class approach. Unlike contrastive learning in the image domain, multivariate time-series anomaly detection presents two clear differences. First, training data are potentially overlapped across time and time steps are shared between them. For example, $\mathbf{W}_t$ and $\mathbf{W}_{t+1}$ share every observation except for one time step. Second, temporal dynamics exist across time steps, and the training data share a common context in that they are continuously operated under normal behavior.

In addition to these differences, we observed that merely using the NT-Xent loss significantly degrades the multivariate time-series anomaly detection performance, as shown in Table II. This observation empirically demonstrates the limitation of the instance discriminating contrastive loss in multivariate time-series anomaly detection. Instance discrimination contrasts views from different window segments; however, every window segment exhibits normal behavior and can exhibit the same temporal dynamics.

Hence, we incorporated contrastive loss with one-class classification for time-series anomaly detection. Because a training batch $\mathbf{W}$ comprises normal segments that share a temporal context, we regarded the training segments as one-class, i.e., the

normal class, and trained an encoder that attracts their representations simultaneously. Hence, we constructed a contrastive loss such that all augmented views from the window segment in $\mathbf{W}$ are regarded as *positive* to each other. However, no *negatives* for contrast exist if we consider every augmented normal segment to be positive.

To generate negative samples for contrast, we introduced a negative data augmentation random variable $\mathcal{A}^-$. Negative data augmentation has been effectively used in the image domain [13], [15], [21], [30]. Window segments augmented via negative augmentation are regarded as synthetic anomalies, and thus their representations are repelled from the anchor representation. In other words, for normal window segments, negative time-series data augmentation generates negative window segments that deviate from the normal distribution. The intentionally generated negative samples function as negatives and are repelled from every normal window segment because every normal window is regarded as positive. In terms of the difference between $\mathcal{A}$ and $\mathcal{A}^-$, both of which are random variables of one of the proposed seven general time-series data augmentation methods, $\mathcal{A}$ aims to generate semantically related samples, whereas $\mathcal{A}^-$ aims to generate synthetic anomalies. The experimental analysis and a guideline for selecting both augmentation variables are introduced in Section IV-F.

Specifically, for a mini-batch $\mathcal{B} = \{\mathbf{W}_t\}_{t=0}^{N-1} \subseteq \mathbf{W}$ comprising N normal window segments, a negative mini-batch $\mathcal{B}^- = \mathcal{T}^-(\mathcal{B})$ is constructed, where $\mathcal{T}^-$ is the realization of $\mathcal{A}^-$. Based on semantic-preserving data augmentations $\mathcal{T}_1, \mathcal{T}_2 \sim \mathcal{A}$ and two types of mini-batches $\mathcal{B}$ and $\mathcal{B}^-$, a multi-viewed mini-batch $\tilde{\mathcal{B}}$ is constructed as follows:

$$\begin{aligned} \tilde{\mathcal{B}} &= \{\tilde{\mathbf{W}}_t\}_{t=0}^{4N-1}, \\ \text{where } \{\tilde{\mathbf{W}}_t\}_{t=0}^{2N-1} &= \bigcup_{i=1,2} \mathcal{T}_i(\mathcal{B}), \\ \{\tilde{\mathbf{W}}_t\}_{t=2N}^{4N-1} &= \bigcup_{i=1,2} \mathcal{T}_i(\mathcal{B}^-). \end{aligned} \quad (11)$$

As mentioned above, our one-class contrastive loss considers instances from the training normal distribution and augmented negative distribution as positives and negatives, respectively. Therefore, for each anchor window segment $2N - 1$ positive and $2N$ negative samples exist in $\tilde{\mathcal{B}}$. The loss function is expressed as follows:

$$\begin{aligned} \mathcal{L}_{\text{oc}} &= -\frac{1}{2N} \sum_{i=0}^{2N-1} \frac{1}{|i^+|} \sum_{k \in i^+} \log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}{\sum_{j \in i^- \cup i^+} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}, \\ \text{where } i^+ &= \{0, \dots, 2N - 1\} \setminus \{i\}, \quad i^- = \{2N, \dots, 4N - 1\}. \end{aligned} \quad (12)$$

Fig. 3 illustrates a comparison between CTAD and other types of contrastive learning frameworks. SimCLR generates instances from only the training data and discriminates instances at the instance level. CSI uses the concept of negative augmentation to generate distributionally shifted instances but discriminates instances at the instance level. Although a supervised variant of CSI exists, it is only applicable in a labeled multi-class setting.

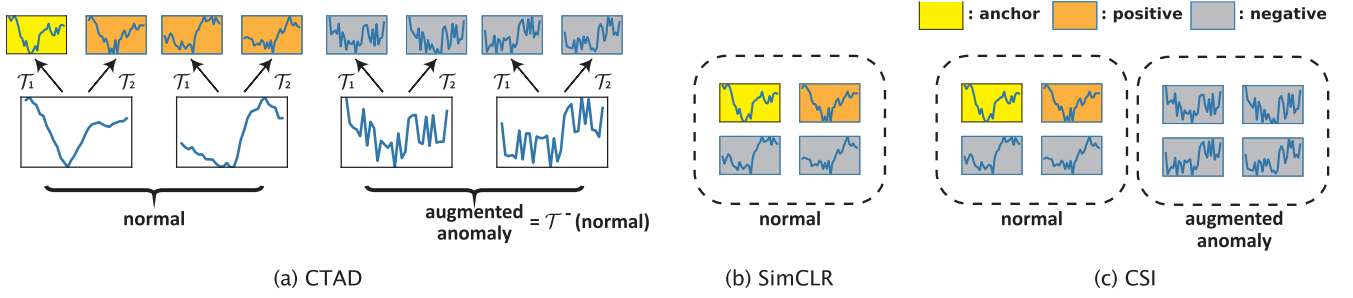


Fig. 3. Comparison of the one-class contrastive learning framework of CTAD with SimCLR and CSI.

The proposed CTAD discriminates instances at the class level, where negative samples are generated solely through negative data augmentation.

E. Anomaly Score

By contrasting the generated anomalous window segments against the normal window segments, we expect unseen anomalous window segments to be clustered far from the normal cluster. Hence, the distance between the centroid of the normal cluster and the embedding of the unseen window sample can be considered as the extent to which the unseen window sample deviates from the normal behavior. We defined the anomaly score $\mathcal{S}$ of unseen window $\mathbf{W}'_t$ as follows:

$$\mathcal{S}(\mathbf{W}'_t; f_\theta, \mathbf{z}_{\text{cent}}) = \mathcal{D}(f_\theta(\mathbf{W}'_t), \mathbf{z}_{\text{cent}}), \quad (13)$$

$$\mathbf{z}_{\text{cent}} = \frac{1}{T - s + 1} \sum_{i=1}^{T-s+1} f_\theta(\mathbf{W}_i) \quad (14)$$

where $\mathbf{z}_{\text{cent}}$ is the centroid of the embeddings of training batch $\mathbf{W}$ and $\mathcal{D}(\cdot, \cdot)$ can be any type of the distance function such as the L_2 distance or cosine distance. Because $\mathbf{z}_{\text{cent}}$ can be pre-computed and stored after the training ends, fast inference is enabled by computing only $f_\theta(\mathbf{W}'_t)$ during the inference time.

IV. EXPERIMENTS

A. Dataset Description

We conducted experiments on four widely used public datasets: Secure Water Treatment (SWaT) [31], Water DIstribution (WADI) [32], Mars Science Laboratory rover (MSL) [33], and Soil Moisture Active Passive satellite (SMAP) [33].

SWaT¹ represents a scaled-down version of the real-world industry water treatment plant. Each time step consisted of values obtained from 51 sensors and actuators. 11 days of continuous operations are split into training and test sets. For the first 7 days, the system is under normal operation (training set). During the following 4 days, 41 cyber and physical attacks are launched (test set).

WADI testbed is an extension of the SWaT testbed. It consists of 16 days of continuous operation. The system is in normal

TABLE III
STATISTICS OF THE EXPERIMENTAL DATASETS

Dataset	Variables	Train set	Test set	Anomaly (%)
SWaT	51	495,000	449,919	12.13
WADI	123	784,537	172,801	5.77
MSL	55	58,317	73,729	10.27
SMAP	25	135,183	417,617	13.13

operation during the first 14 days, and 15 different attacks are launched in the subsequent 2 days. The values of 123 sensors and actuators are obtained during this period. The data characteristics of the datasets are summarized in Table III.

MSL and **SMAP** are spacecraft telemetry datasets from NASA. They contain multivariate data captured from multiple telemetry channels across multiple entities. The MSL and SMAP datasets contain 27 and 55 entities, respectively, and each entity contains 55 and 25 telemetry channels, respectively. Similar to SWaT and WADI, the training data contain behaviors under normal operations. Therefore, spacecraft anomalies exist only in the test data, which are labeled by expertise.

B. Model Architecture and Hyper-Parameters

The encoder network f_θ consists of one LSTM layer, followed by a projection layer. The projection layer is known to be important for contrastive learning [19], and we used the sequential operation of MLP - Batch Normalization - ReLU - MLP layers as the projection layer. The hidden state dimension of the LSTM was set to 64 for SWaT and 128 for WADI, and the hidden state vector of the last time step was used as the input to the projection layer. Both MLP layers in the projection layer had output dimensions of 128. Using this relatively simple network, we show that CTAD consistently performs well on various datasets. We optimized the network using the Adam optimizer [34] with a learning rate of 0.001 and weight decay of 0.0001. The learning rate gradually decayed to zero as the training progressed, which involved the use of a cosine annealing learning rate scheduler [35]. The network was trained for 100 epochs with a batch size of 1024 for SWaT, and 50 epochs with the same batch size for WADI. The temperature parameter τ was set to 0.1 in all experiments. For data augmentation, $\mathcal{T}_{\text{Gaussian}}$ was used as augmentation $\mathcal{A}$ and $\mathcal{T}_{\text{bias}}$ was used as negative augmentation $\mathcal{A}^-$. Detailed hyper-parameters of the window

¹We used version 1 dataset. In version 1, the first 30 minutes of data are removed from version 0 which is not considered a normal operation.

TABLE IV
HYPER-PARAMETER SETTINGS FOR WINDOW SIZE, DATA AUGMENTATION, AND DISTANCE FUNCTION

	window size	amplify			bias			dropout		reverse		permute		mask		noise		distance
	s	a	p	n_v	b	p	n_v	p	n_v	p	n_v	p	n_v	p	n_v	σ	n_v	$\mathcal{D}$
SWaT	4	1.5	0.5	1	0.2	0.5	1	0.5	25	0.5	25	0.5	25	0.5	25	0.05	51	L_2
WADI	4	1.5	0.5	1	0.1	0.5	1	0.5	61	0.5	61	0.5	61	0.5	61	0.02	123	cosine
MSL	4	1.5	0.5	1	0.2	0.5	1	0.5	27	0.5	27	0.5	27	0.5	27	0.05	55	L_2/cosine
SMAP	4	1.5	0.5	1	0.2	0.5	1	0.5	12	0.5	12	0.5	12	0.5	12	0.05	25	L_2/cosine

size, data augmentations, and distance function are presented in Table IV.

C. Compared Anomaly Detection Methods

We compared our method with i) classical methods, ii) existing multivariate time-series anomaly detection methods, iii) a one-class anomaly detection method, and iv) self-supervised learning methods.

The compared classical methods include autoencoders (AE) and Isolation Forests (IF). These methods can be summarized as:

- The AE trains the encoder–decoder architecture that reconstructs the training data, and the reconstruction error of the test data is defined as an anomaly score.
- The IF [36] isolates data in the form of a decision tree. The anomaly score is defined based on the path length of the tree.

The compared multivariate time-series anomaly detection methods can be summarized as follows:

- The LSTM-VAE [37] constructs a VAE using LSTM layers and a progress-based prior. The anomaly score is defined based on the reconstruction probability of the data.
- DAGMM [38] learns the approximated density distribution of the data by jointly optimizing a deep AE and a Gaussian mixture model to evaluate the likelihood of the samples.
- OmniAnomaly [39] renders the VAE representation more robust via a stochastic variable connection and a planar normalizing flow.
- USAD [40] adversarially trains the AE in two phases. In Phase 1, the standard AE is trained, and in Phase 2, the reconstructed output is used as an input for the second AE.

The compared one-class anomaly detection method includes one representative method, and we implemented an encoder network similar to that for CTAD without using a projection layer. The method is summarized as follows:

- Deep SVDD [41] minimizes the volume of a hypersphere that encloses the training data representation. The anomaly score is defined as the distance between the test representation and the center of the hypersphere.

Self-supervised learning methods include contrastive and non-contrastive learning methods. These methods were initially proposed in the image domain and thus used data augmentations for the image. We implemented these methods using the proposed time-series data augmentation and encoder architecture same as CTAD. The self-supervised learning methods are briefly summarized as follows:

- SimCLR [19] uses randomized data augmentation to generate positive and negative samples and trains an encoder to predict a positive sample.
- CSI [13] additionally generates negative samples via negative data augmentation using the SimCLR contrastive loss. We exclude the auxiliary classifier of the CSI to perform an unbiased comparison of the effectiveness of contrastive loss.
- MoCo [23] uses the momentum encoder and dictionary queue to increase the number of negative samples. The representations of the negative samples are stored in a queue.
- BYOL [42] is a non-contrastive learning method in which the encoder is trained to predict the representation of the augmented view from the same image.

For the comparison with classical methods and existing time-series anomaly detection methods, the results in [40] were referred to. It should be noted that we performed the same pre-processing as that reported in [40] for an unbiased comparison.

D. Experimental Setup

In all the experiments of the CTAD and compared methods, the training set is min-max normalized in the range [0, 1] before constructing the window-sliced batch $\mathbf{W}$. Every test set is normalized with the min and max values of the corresponding training set, i.e., $\mathbf{x}_{i,\text{normalized}} = (\mathbf{x}_i - \min_{\text{train}}) / (\max_{\text{train}} - \min_{\text{train}})$, $1 \leq i \leq T'$. It should be noted that normalizing the test set with the min and max values of the test set can result in a mismatch between the training and test sets (e.g., the overall scale of the signal can be reduced by an unexpectedly high value of anomalous observation). We downsampled all datasets at a rate of 5 before normalization using the median values, following the setting provided in [40]. When the test data had less than 1,200 time-steps for MSL and SMAP, we did not downsample the data because downsampling erased anomalous observations.

For the evaluation metrics, we used the Area Under the Receiver Operating Characteristic (AUROC), Area Under the Precision-Recall curve (AUPR), and F1-score (F1). Most multivariate time-series anomaly detection methods use only F1 to evaluate the detection performance. Additionally, we report the threshold-independent metrics (e.g., AUROC and AUPR), which can be used to evaluate the model performance without the effect of threshold selection.

In [43], F1 with point-adjust (PA) was proposed because anomalous observations typically appear in segments (continuously in time) and correctly detecting at least once in the segment

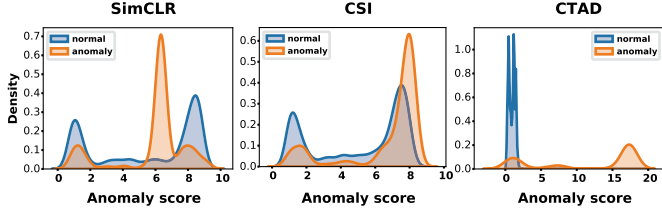


Fig. 4. Conditional probability distribution of anomaly score given normal observation $p(S|y' = 0)$ and anomalous observation $p(S|y' = 1)$ (best viewed in color).

is sufficient. Specifically, for each ground-truth anomalous segment, if any point in the segment is correctly detected as an anomaly, then the entire segment is adjusted to be correctly detected as an anomaly. Additionally, we adjusted F1 with PA based on previous studies and reported the value herein. F1, precision, and recall adjusted with PA are denoted herein by $F1_{PA}$, $precision_{PA}$, and $recall_{PA}$, respectively. We report the averaged AUROC and AUPR over the entities for MSL and SMAP, which contain multiple entities. Similarly, we calculated F1 and $F1_{PA}$ based on the averaged precision, recall, $precision_{PA}$, and $recall_{PA}$ based on the setting provided in [39].

All experiments were performed on a server with the following specifications: Ubuntu 20.04 OS, Intel i7-9800X CPU with 128 GB of RAM, NVIDIA GeForce GTX Titan X Pascal GPU with 12 GB of VRAM, Python version 3.8.8, PyTorch version 1.7.1, and CUDA version 11.0.

E. Anomaly Detection Performance

Table V presents the anomaly detection performance of the proposed method compared with those of other methods. For the SWaT, WADI, and MSL datasets, CTAD exhibited the highest performance in terms of the AUROC, AUPR, F1, and $F1_{PA}$. MoCo showed the second-best performance among contrastive learning methods. This may be due to the momentum encoder and dictionary queue. The enqueued normal embeddings at the head of the queue are from the old version of the encoder. Thus, they are more suitable as negative samples in the instance discrimination setting. For BYOL, which is a non-contrastive learning method that does not require negative samples, the anomaly detection performance is low for the SWaT, WADI, and MSL datasets.

The effectiveness of one-class contrastive loss is highlighted in the comparison with SimCLR and CSI. Compared to SimCLR, CSI uses negative augmentation to generate additional negative samples for contrast. Although this improves the anomaly detection performance compared with SimCLR (i.e., AUPR from 0.127 to 0.384, and F1 score from 0.279 to 0.455 in SWaT), the performance remained worse than the other multivariate time-series anomaly detection methods. Leveraged by the one-class contrastive loss, CTAD improves the performance compared with CSI (i.e., AUPR from 0.384 to 0.802 ($2.08\times$ higher), and F1 score from 0.455 to 0.841 ($1.85\times$ higher)).

Fig. 4 shows the conditional probability distribution of the anomaly scores of SimCLR, CSI, and CTAD. In CTAD, the

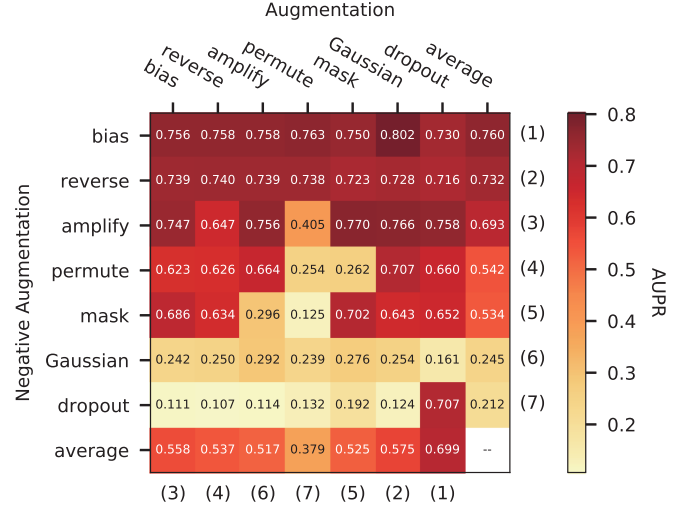


Fig. 5. AUPR for augmentation compositions.

score distribution differs significantly between the normal and anomalies, compared with those of SimCLR and CSI. Moreover, the score distribution of the normal samples was concentrated around 0, thus forming a sharp distribution. This implies that the embeddings of the unseen normal samples were clustered around the centroid of the normal training samples. Moreover, the anomaly score scale was extremely large (from 15 to 20) for the anomalous samples in CTAD. This shows the discriminative ability of CTAD, and the embeddings of the anomalies are clustered far from the centroid of the training normal samples.

F. Effect of Data Augmentation Composition

Fig. 5 shows the AUPR for each data augmentation composition using the SWaT dataset. For the seven designed data augmentation methods, each row and column represents the data augmentation used as negative augmentation $\mathcal{A}^-$ and augmentation $\mathcal{A}$, respectively. The last row and column represent the averaged AUPR for each augmentation and negative augmentation, respectively. The numbers in parentheses indicate the average AUPR rankings of each augmentation, when used as $\mathcal{A}$ or $\mathcal{A}^-$. The results were sorted based on the ranking of $\mathcal{A}^-$.

We obtained three sets of important results. First, the anomaly detection performance was primarily affected by the type of negative augmentation instead of the semantic-preserving augmentation. For example, when the negative augmentation $\mathcal{A}^-$ is $\mathcal{T}_{bias}$ or $\mathcal{T}_{reverse}$, the AUPR is consistently high regardless of $\mathcal{A}$. In the case of $\mathcal{T}_{Gaussian}$ or $\mathcal{T}_{dropout}$, however, the AUPR is consistently low when they are used as negative augmentation. Second, the augmentation that is not suitable for negative augmentation $\mathcal{A}^-$ is suitable for augmentation $\mathcal{A}$. In fact, $\mathcal{T}_{Gaussian}$ and $\mathcal{T}_{dropout}$ are the two worst alternatives for negative augmentation, although they are the two best ones for augmentation. Finally, scale-perturbing augmentation is typically preferred for negative augmentation; however, perturbing to zero significantly deteriorates the performance. Meanwhile, both $\mathcal{T}_{bias}$ and $\mathcal{T}_{amplify}$ are high-performing negative augmentation alternatives, whereas

TABLE V
PERFORMANCE COMPARISON WITH COMPARED METHODS

Model	SWaT							
	AUROC	AUPR	Precision	Recall	F1	Precision _{PA}	Recall _{PA}	F1 _{PA}
AE [†]	–	–	0.990	0.630	0.770	0.991	0.704	0.823
IF [†]	–	–	0.951	0.588	0.727	0.962	0.732	0.831
LSTM-VAE [†]	–	–	0.990	0.638	0.776	0.712	0.926	0.805
DAGGM [†]	–	–	0.470	0.666	0.551	0.829	0.767	0.797
OmniAnomaly [†]	–	–	0.983	0.650	0.782	0.722	0.983	0.833
USAD [†]	–	–	0.985	0.662	0.792	0.987	0.740	0.846
Deep SVDD	0.839	0.637	0.888	0.547	0.677	0.975	0.726	0.832
SimCLR	0.465	0.127	0.168	0.812	0.279	0.233	0.767	0.357
MoCo	0.804	0.705	0.989	0.595	0.743	0.994	0.677	0.806
BYOL	0.473	0.106	0.171	0.810	0.283	0.209	0.785	0.330
CSI	0.741	0.384	0.510	0.410	0.455	0.895	0.694	0.782
CTAD (ours)	0.880	0.802	0.971	0.742	0.841	0.971	0.881	0.924

Model	WADI							
	AUROC	AUPR	Precision	Recall	F1	Precision _{PA}	Recall _{PA}	F1 _{PA}
AE [†]	–	–	0.995	0.131	0.232	0.397	0.322	0.356
IF [†]	–	–	0.299	0.158	0.207	0.624	0.616	0.620
LSTM-VAE [†]	–	–	0.995	0.128	0.227	0.463	0.322	0.380
DAGGM [†]	–	–	0.065	0.913	0.122	0.223	0.198	0.209
OmniAnomaly [†]	–	–	0.995	0.130	0.230	0.265	0.980	0.417
USAD [†]	–	–	0.995	0.132	0.233	0.645	0.322	0.430
Deep SVDD	0.524	0.214	0.987	0.148	0.257	0.962	0.243	0.388
SimCLR	0.489	0.057	0.073	0.458	0.125	0.680	0.520	0.590
MoCo	0.597	0.073	0.089	0.561	0.154	0.557	0.384	0.455
BYOL	0.396	0.044	0.059	0.875	0.110	0.473	0.260	0.335
CSI	0.489	0.054	0.065	0.553	0.116	0.381	0.384	0.382
CTAD (ours)	0.623	0.399	0.792	0.346	0.481	0.808	0.546	0.652

Model	MSL							
	AUROC	AUPR	Precision	Recall	F1	Precision _{PA}	Recall _{PA}	F1 _{PA}
AE [†]	–	–	–	–	–	0.854	0.975	0.910
IF [†]	–	–	–	–	–	0.568	0.674	0.617
LSTM-VAE [†]	–	–	–	–	–	0.860	0.976	0.914
DAGGM [†]	–	–	–	–	–	0.756	0.980	0.854
OmniAnomaly [†]	–	–	–	–	–	0.914	0.889	0.901
USAD [†]	–	–	–	–	–	0.881	0.979	0.927
Deep SVDD	0.677	0.350	0.487	0.647	0.556	0.866	0.975	0.917
SimCLR	0.670	0.335	0.454	0.624	0.526	0.849	1.0	0.918
MoCo	0.704	0.350	0.478	0.590	0.528	0.860	1.0	0.925
BYOL	0.632	0.336	0.459	0.598	0.519	0.833	1.0	0.909
CSI	0.680	0.351	0.478	0.612	0.537	0.863	1.0	0.926
CTAD (ours)	0.711	0.423	0.575	0.578	0.576	0.952	0.979	0.965

Model	SMAP							
	AUROC	AUPR	Precision	Recall	F1	Precision _{PA}	Recall _{PA}	F1 _{PA}
AE [†]	–	–	–	–	–	0.722	0.980	0.831
IF [†]	–	–	–	–	–	0.442	0.511	0.474
LSTM-VAE [†]	–	–	–	–	–	0.716	0.988	0.830
DAGGM [†]	–	–	–	–	–	0.633	0.998	0.775
OmniAnomaly [†]	–	–	–	–	–	0.759	0.976	0.854
USAD [†]	–	–	–	–	–	0.770	0.983	0.863
Deep SVDD	0.607	0.303	0.477	0.583	0.525	0.781	0.976	0.868
SimCLR	0.614	0.311	0.408	0.605	0.487	0.825	0.972	0.892
MoCo	0.611	0.273	0.377	0.622	0.469	0.770	0.972	0.859
BYOL	0.654	0.301	0.426	0.620	0.505	0.782	0.991	0.874
CSI	0.615	0.307	0.429	0.564	0.487	0.795	0.972	0.875
CTAD (ours)	0.639	0.323	0.492	0.570	0.528	0.830	0.972	0.895

[†] Referenced from the results of [40]

TABLE VI
ABLATION STUDIES ON COMPONENTS OF CTAD

	BN	Pos. aug.	Sum out	Schedule	AUPR
w/o BN	✗	✓	✓	✓	0.121
w/o Pos. aug.	✓	✗	✓	✓	0.262
Sum in	✓	✓	✗	✓	0.384
w/o Schedule	✓	✓	✓	✗	0.772
CTAD	✓	✓	✓	✓	0.802

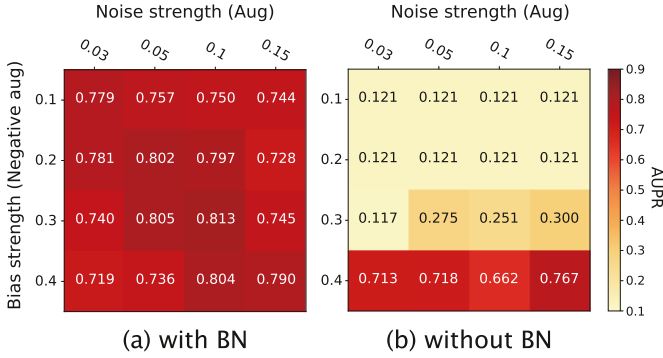


Fig. 6. AUPR for different augmentation strengths with and without batch normalization (BN). Bias strength and noise strength are represented by b and σ , respectively. (a) With BN, AUPR increased for weak negative augmentation. (b) Without BN, weak negative augmentation resulted in mode collapse.

$\mathcal{T}_{\text{dropout}}$ exhibits the worst performance. This is because perturbing to zero removes the information. $\mathcal{T}_{\text{mask}}$ exhibits inferior performance, with zeros in the time steps. In summary, the type of negative augmentation is the most important factor in contrastive time-series data augmentation, and $\mathcal{T}_{\text{bias}}$ is generally a good option. These observations are crucial when designing and selecting time-series data augmentation methods for contrastive time-series anomaly detection.

G. Ablation Studies

We performed experimental analyses to determine components that are vital to CTAD. Table VI summarizes the AUPR when each component of CTAD is eliminated. Most notably, batch normalization (BN) is the key difference. When BN is removed from the projection layer, the learned embeddings collapse to a single point, resulting in a significant decrease in the AUPR from 0.802 to 0.121.

Fig. 6 further analyzes this phenomenon. Herein, we report the AUPR for different augmentation and negative augmentation strengths. Additive bias is used as a negative augmentation $\mathcal{A}^-$ and the bias scale b is increased from 0.1 to 0.4 by 0.1. Additive Gaussian noise is used as an augmentation $\mathcal{A}$ and the used standard deviations σ are 0.03, 0.05, 0.1, and 0.15. When the strength of the negative augmentation is weak, negative augmentation generates hard anomalous samples that are difficult to distinguish from normal samples. If BN is not used, then the encoder network cannot distinguish the hard anomalous samples from normal samples. Therefore, the embeddings are collapsed to a single point for the weak negative augmentation

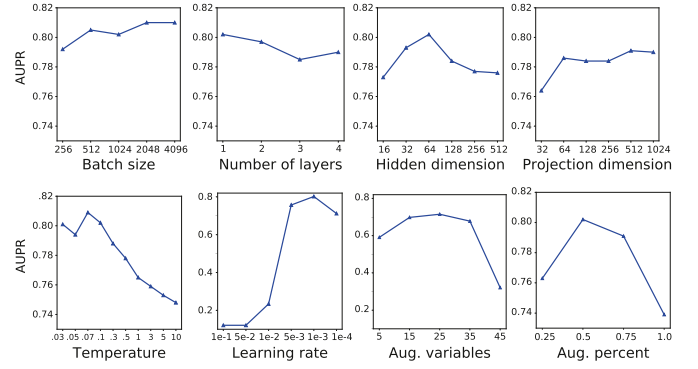


Fig. 7. Ablation studies on various components of the CTAD such as batch size, model size, temperature, learning rate, and augmentation parameters.

strengths. BN appears to prevent collapse and thus enables the utilization of near anomalous samples. The effective use of hard negative samples allows the contrastive learning encoder to be trained more effectively [44], [45], and the use of hard anomalous samples is also effective for time-series anomaly detection.

Because CTAD considers every normal window segment as positive, multiple positives exist even without augmentation $\mathcal{A}$ (i.e., $N - 1$ positives which are different normal window segments in the mini-batch). We denote augmentation $\mathcal{A}$ as positive augmentation, and Table VI shows that the AUPR decreases from 0.802 to 0.262 for the case without positive augmentation. This implies that positive augmentation is important in CTAD. Moreover, as each anchor contains $2N - 1$ positive samples, the exponential terms from the positive samples can be summed inside the logarithm as:

$$\mathcal{L}_{\text{oc}} = -\frac{1}{2N} \sum_{i=0}^{2N-1} \log \left(\frac{1}{|i^+|} \cdot \frac{\sum_{k \in i^+} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}{\sum_{j \in i^- \cup i^+} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)} \right). \quad (15)$$

When (15) is used as a one-class contrastive loss, the AUPR deteriorated to 0.384, as shown in Table VI. This tendency was first observed in contrastive representation learning with supervision in the image domain [18]; similarly, this is also applicable to CTAD.

Cosine learning rate scheduling is another important factor that improves contrastive representation learning [19]. We investigated whether this is applicable to multivariate time-series anomaly detection. Table VI shows that the absence of cosine learning rate scheduling decreased the AUPR from 0.802 to 0.772. In other words, cosine scheduling enhanced anomaly detection performance, but the effect was insignificant compared with those of other components.

We conducted ablation studies by changing the hyperparameters of several components of CTAD, such as the batch size, number of LSTM layers, LSTM hidden state dimension, projection layer dimension, temperature, learning rate, and augmentation parameters. The results are shown in Fig. 7. When the

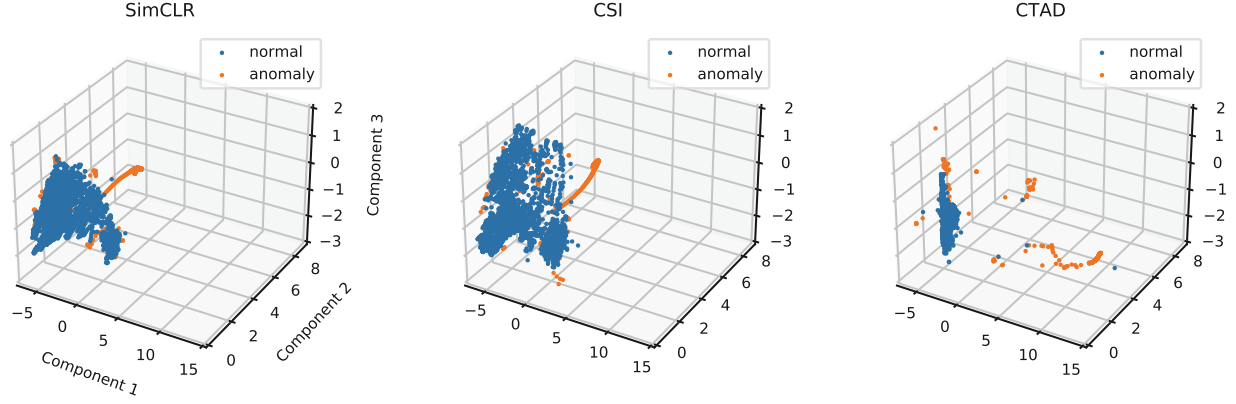


Fig. 8. Visualization of PCA results of SimCLR (left), CSI (middle), and CTAD (right).

number of layers was increased, the performance deteriorated slightly but remained similar in general. This supports the idea that simple networks are sufficient for achieving good performance in multivariate time-series anomaly detection. Existing methods may be over-parameterized to fit multivariate time-series datasets. The most appropriate hidden state dimension of LSTM is 64, and a higher dimension degrades the performance. This shows that a higher model capacity is not the best choice. The performance improves when the number of variables to be augmented is randomly selected (line 2 of Algorithm 1). Moreover, instead of augmenting the entire time-series of the selected variables, providing additional randomness by augmenting only a few percent of the time-series (line 5 of Algorithm 1) is more effective. This is because the increased randomness provides more variety to the generated samples and augmenting a small portion, instead of the entire sample, can generate hard anomalies.

To summarize the important implications of the ablation studies, BN is vital to the successful anomaly detection of CTAD, and using various samples and hard-negative samples for contrast is crucial. Thus, increasing randomness and generating hard negative samples using the proposed augmentation strategy improves the anomaly detection performance of CTAD.

H. Representation Separateness

To verify whether the normal and anomalous window segments were separated sufficiently in the embedding space, we visualized the embeddings of the test data via principal component analysis in a three-dimensional space. Fig. 8 shows the comparison of the visualizations of SimCLR, CSI, and CTAD. In CTAD, anomalous segments were well clustered and separated from normal segments, and the embeddings of the normal segments were highly clustered. Conversely, the embeddings of SimCLR and CSI were not well separated between normal and anomalous segments. Meanwhile, in CTAD, the scale of most principal components (Component 1) was much larger for the anomalous segments compared with the other methods. This implies that CTAD performs well because the principal component represents the degree of the anomaly.

TABLE VII
MODEL SIZE AND INFERENCE TIME

	Model size (KB)	Inference time (s)
OmniAnomaly	10,313 (47.96 $\times$)	18.902 (20.21 $\times$)
USAD	332 (1.54 $\times$)	1.277 (1.36 $\times$)
CTAD	215 (1.00 $\times$)	0.935 (1.00 $\times$)

I. Model Size and Inference Time

Table VII presents the model size and inference time compared with OmniAnomaly and USAD. Compared with OmniAnomaly, CTAD has 47.96 $\times$ fewer parameters and demonstrates 20.21 $\times$ faster inference. Compared with USAD, CTAD has 1.54 $\times$ fewer parameters and demonstrates 1.36 $\times$ faster inference. CTAD demonstrates fast and memory-efficient performance, which is vital to time- and memory-critical tasks such as real-time monitoring or anomaly detection in wearable devices.

J. Noise Robustness

By incorporating additive Gaussian noise as the augmentation of the contrastive learning framework, we expect another desirable property of the time-series anomaly detection model: robustness to noise. Specifically, the “noise augmented view” of the window segments are considered positive to each other, resulting in invariance to noise. Robustness to noise is crucial because signals can interfere with undesired noise. This can increase the number of false positives, i.e., noisy normal segments can be predicted as anomalies.

To demonstrate that CTAD performs robustly, we considered two scenarios where the training data are perturbed or the test data are perturbed at a certain ratio. In both cases, perturbation was injected with independent and identically distributed Gaussian noise sampled from $\mathcal{N}(0, (0.3)^2)$. This setting is similar to that presented in [40], however, in [40], noise was injected only into the training data.

Table VIII shows the AUPR as the perturbation rate increased from 0 to 0.3. When CTAD was trained with the perturbed

TABLE VIII
ROBUSTNESS TO NOISE FOR DIFFERENT PERTURBATION RATES

	Perturbation rate					
	0.00	0.01	0.02	0.05	0.10	0.30
AUPR (Train)	0.802	0.796 (-0.7%)	0.794 (-1.0%)	0.789 (-1.6%)	0.729 (-9.1%)	0.726 (-9.5%)
AUPR (Test)	0.802	0.785 (-2.1%)	0.784 (-2.2%)	0.776 (-3.2%)	0.720 (-10.2%)	0.774 (-3.5%)

training data, the AUPR was consistently high with little degradation until a perturbation rate of 0.05. Specifically, the AUPR decreased by only 1.6% at a perturbation rate of 0.05. Similarly, when the test data were perturbed, AUPR decreased only slightly, i.e., 3.2%, at a perturbation rate of 0.05.

V. DISCUSSION

We designed three types of general time-series data augmentation methods, and these methods present the following advantages. First, compared with augmenting the entire time series, the randomness of augmentation is significantly diversified by two controlling parameters n_v and p , which improves the detection performance (bottom right of Fig. 7). Second, scale-perturbing augmentation can generate point-wise anomalies, which can represent hard anomalies in real-world scenarios such as glitching to only a few specific variables. The generated hard anomalies enhance the detection performance when BN is applied, as shown in Fig. 6. Finally, scale-perturbing augmentation can minimize the hurt of temporality because the augmentation does not occur in consecutive periods.

One potential limitation of the current augmentation method is that the augmentation strength relies on pre-defined parameters, such as b or σ . Adaptive time-series augmentation strategies, such as determining the augmentation strength based on the signal power of the window or generating synthetic anomalies via generative models, instead of negative augmentation, would be an interesting future direction. Moreover, whereas our one-class contrastive loss considers every window segment in the training data to be positive, more closely related window segments may exist, e.g., window segments that are closer in time can share a more temporal context. We expect the positive sample mining strategy, which quantifies the temporal context between window segments, to further enhance multivariate time-series anomaly detection performance. In addition, the application of our results to other applications, such as anomaly detection in embedded systems that require memory-critical service as well as the anomaly detection of sequential data, would be an interesting future investigation.

The model performs well even under a moderate amount of noise because it is trained in similar situations, i.e., noisy window segments are regarded as positives. Having prior knowledge regarding the potential noise would allow one to adjust the robustness of the model by setting the appropriate noise augmentation strength during training.

VI. CONCLUSION

Herein, we proposed a novel one-class contrastive learning method that considers all data in the training data distribution as

positive samples and uses negative samples solely from negative distributions generated via negative data augmentations. Moreover, we designed general time-series data augmentations for time-series contrastive learning and empirically demonstrated the effectiveness of the proposed data augmentations for time-series anomaly detection. We identified the key role of BN in enabling the exploitation of hard anomalous samples. The simple yet effective modification of the contrastive loss and the appropriate time-series data augmentation method enable accurate, fast, memory-efficient, and robust multivariate time-series anomaly detection. In future investigations, one-class contrastive loss should be applied to other domain tasks and the role of BN in contrastive anomaly detection should be analyzed theoretically.

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